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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

EC4352-1 – ADVANCED CYBER SECURITY
(For ECE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **One** full question from each **Unit**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which is following element is a primitive element in the group $G = \{1, 2, 3, 4, 5, 6\}$ over mod 7 multiplication?
A) 2
B) 3
C) 4
D) 6
2. The value of $7^{147} \bmod 13$ is
A) 12
B) 8
C) 11
D) 5
3. Which of the following is an example of a pervasive security mechanism?
A) Security Label
B) Authentication Exchange
C) Traffic Padding
D) Notarization
4. Number of rounds in the decryption process of DES is
A) 8
B) 16
C) 64
D) 32
5. Which of the following properties is not a mandatory property for a Group?
A) Associative
B) Existence of identity element
C) Commutative
D) Closure
6. Which of the following is called as Permutation Ciphers?
A) Caesar Cipher
B) Hill Cipher
C) Playfair Cipher
D) Transposition Cipher
7. The attack that involves reordering of the legitimate message is called
A) Masquerade
B) Replay
C) Modification of messages
D) Denial of service
8. Find the value of $8^{147} \bmod 13$ is
A) 8
B) 5
C) 11
D) 7
9. The prime numbers considered in an RSA cryptosystem are 11 and 5. The corresponding $\phi(n)$ is
A) 55
B) 44
C) 50
D) 40
10. A digital signature needs which of the following?
A) Private-key system
B) Shared-key system
C) Public-key system
D) It does not require a key
11. Which of the following is not a valid pair of (p, q) in an RSA cryptosystem?
A) (5, 11)
B) (15, 21)
C) (11, 13)
D) (17, 29)
12. Which of the following is not a method to distribute public keys?
A) Digest
B) Using public-key certificates
C) Authority
D) Directory

13. Using public key cryptography, X adds a digital signature σ to message M, encrypts $\langle M, \sigma \rangle$, and sends it to Y, where it is decrypted. Which one of the following sequences of keys is used for the operations?
- A) Encryption: X's private key followed by Y's private key; Decryption: Y's public key followed by X's public key
 B) Encryption: X's private key followed by Y's public key; Decryption: Y's public key followed by X's private key
 C) Encryption: X's public key followed by Y's private key; Decryption: Y's public key followed by X's private key
 D) Encryption: X's private key followed by Y's public key; Decryption: Y's private key followed by X's public key
14. Output length of Message Authentication Code is
- A) Dependent on the length of the message only
 B) Dependent on the length of the key only
 C) Fixed
 D) Dependent on both the length of the message and the length of the key
15. Which of the following statements is TRUE?
- I. Message authentication protects two parties who exchange messages from any third party
 II. Message authentication can be used in the situations where there is not a complete trust between sender and receiver
- A) Neither I Nor II
 B) I Only
 C) Both I and II
 D) II Only
16. In the RSA algorithm, we select 2 random large values 'p' and 'q'. Which of the following is the property of 'p' and 'q'?
- A) p and q should be prime
 B) p and q should be divisible by $\Phi(n)$
 C) p and q should be even numbers
 D) p/q should give no remainder
17. What is the purpose of a sandbox in the context of cybersecurity?
- A) Displaying unwanted advertisements
 B) Encrypting files for ransom
 C) Spreading across networks
 D) Testing and isolating potentially malicious code
18. Which of the following is not a technique used in a phishing attack?
- A) Social Engineering
 B) Creating a sense of urgency
 C) Distributed Denial of Service
 D) Fake Website
19. What is the purpose of Rootkit?
- A) Stealing sensitive information
 B) Concealing unauthorized access
 C) Encrypting files for ransom
 D) Disrupting network services
20. This is a type of Trojan that enables the attacker to gain access to and execute commands on the compromised system is called
- A) Backdoor
 B) Maldoor
 C) Sidedoor
 D) Frontdoor

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Make use of Extended Euclid algorithm to find Multiplicative Inverse of 550 mod 1759.	08	L*3	1	1
	b) Explain the operations involved in a single round of DES Algorithm.	08	L2	2	1
2.	a) State Fermat's Little theorem. Also calculate $9^{1058} \bmod 97$.	08	L3	1	1
	b) Explain AES Algorithm in detail.	08	L2	2	1
3.	a) Solve for the least non-negative integer x that satisfies $x \equiv 5 \pmod{7}$ $x \equiv 2 \pmod{9}$ $x \equiv 7 \pmod{10}$	08	L3	1	1
	b) Explain the process of encryption and decryption using play fair cipher. Find the cipher text using the key "NMAMIT" and the plain text "electronics and communication".	08	L3	2	1

Unit – II

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|-------|--|----|----|---|---|
| 4. a) | Describe the encryption and decryption process in Elliptic curve cryptography. Also explain the analogy of ECC Diffie Hellman Key Exchange. | 08 | L2 | 3 | 1 |
| b) | Explain in detail, the types of Digital signatures. | 08 | L2 | 4 | 1 |
| 5. a) | Explain RSA Algorithm. Determine cipher value if the following key values are used for encryption prime numbers $p=17$, $q=31$, $e=7$ and $M=2$. | 08 | L3 | 3 | 1 |
| b) | Give the requirements of message authentication codes. Also mention the basic uses of Message Authentication Code. | 08 | L2 | 4 | 1 |
| 6. a) | Explain Diffie Hellman Key exchange algorithm. Show how User A and B calculates same secret key using the values $q=353$, $\alpha=3$, $X_A=97$ and $X_B=233$. | 08 | L3 | 3 | 1 |
| b) | With the help of neat block diagrams explain the variants of Hash function. | 08 | L2 | 4 | 1 |

Unit – III

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|-------|---|----|----|---|---|
| 7. a) | What is malware fingerprinting? Provide a detailed explanation covering goals, common indicators, and how fingerprints support detection, attribution, and threat intelligence. | 08 | L2 | 5 | 1 |
| b) | In the context of static analysis, explain the techniques for determining a file's type and format. | 08 | L2 | 5 | 1 |
| 8. a) | Define malware analysis and justify its necessity in modern cybersecurity operations. | 08 | L2 | 5 | 1 |
| b) | Describe any four categories of malware in detail, explaining their core mechanisms, propagation strategies, and typical impact on target systems, with suitable examples. | 08 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
